QUICK CARD · COMPOSITES LAB (KOMPOSITTLABBEN)

# Stepcraft M700 – CNC mill

UCCNC · SolidWorks CAM · § = section in the user manual

Full user manual with photos: **scan the QR code** or go to **ravo10.github.io/USN-kongsberg-lab-mech**.

**SAFETY**

- **Do not use the machine** without training
- Clamps **≥ 50 mm** outside the milling area
- **Emergency stops** outside and inside the door
- Use at most **640 × 480 mm** of the table

## A. Start-up §3.01

1. Turn on the **compressed air**.
2. Turn on the **vacuum** (switch above the mill).
3. Release the **emergency stops** (outside and inside).
4. Start **UCCNC** on the PC.
5. Turn on the **spindle** (flashes blue/red).
6. Click **Reset** bottom right.

## B. Z zero point with the TLS §4.07

7. Clamp the stock.
8. **M6** → the tool used first.
9. Move above one of the **highest points** of the stock.
10. Put the **TLS** on the stock, under the tool.
11. **Page Up/Down** until just above the TLS (not touching).
12. Press **TLS** in UCCNC and wait.

Do this every time the stock has a new thickness. X/Y: zero in UCCNC or find the origin with the 3D Touch Probe.

## Troubleshooting §3.07, §4.05

| | |
|---|---|
| **Wrong toolpath in UCCNC** | Close and reopen UCCNC |
| **Tool will not release** | Chuck lock + spanner (§4.05) |
| **Zero point lost** | Find the origin again with the 3D Touch Probe |
| **Holes < Ø6 mm** | Mill Ø2 mm, drill out by hand afterwards |

## C. Tools and magazine §4.01–4.05

- **DOUBLE HOME ALL** → **OFFSETS** to calibrate the tools.
- Tools **1–8**. No. 9 is disabled, no. **10 = 3D Touch Probe**. Change with **M6** + no.
- Tool 1 is the Z reference for the probe.
- Stuck tool: press in the **chuck lock**, turn with the spanner.
- Keep the chuck clean: **vacuum** it during service.

## D. DXF → G-code in UCCNC §3.05

13. DXF layers: no. 1 = **0**, no. 2 = **11**, then **12, 13 ...** (clean up in QCAD).
14. **CAM** tab → **Import DXF File** → choose the layer.
15. Choose the **origin** and the **Operation type** (inside/outside).
16. **Generate Toolpath** → **Create G-code**, one layer at a time.

**Example (Ø2 mm cutter)**

| | |
|---|---|
| **Tool diameter** | 2 |
| **Final depth** | -6 |
| **Cut per pass** | 2 |
| **Feedrate / plunge** | 235 / 80 |
| **Spindle speed** | 23000 |
| **Safe Z** | 10 |

## E. For students (SolidWorks) §2

- Origin at the centre of a **circle or rectangle** (the probe can find it again).
- **Z up**, X/Y as the part lies in the machine.
- The part must fit inside **640 × 480 mm** minus the clamps.
- **Pocket** (Finish) recommended, with tabs if needed.
- Holes under Ø6 mm: separate sketch with **Ø2 mm** holes.